Notes on MATH 223

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Chapter 1. Complex Numbers

1. Structure of Complex Numbers

Definition 1.1.1 (Complex Numbers):

The set of **complex numbers**, denoted by $\mathbb{C}$, is the Cartesian product of the real numbers with itself. That is

$$\mathbb{C} := \mathbb{R} \times \mathbb{R}.$$

A typical element $z \in \mathbb{C}$ can be expressed as an ordered pair of real numbers (x, y). The set of complex numbers as for now is just a set of ordered pairs. Thus, we shall define some operations on $\mathbb{C}$ to give it more structure.

Definition 1.1.2 (Operations Within Complex Numbers):

The **addition** of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is defined as

$$z_1 + z_2 := (x_1 + x_2, y_1 + y_2).$$

The **multiplication** of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is defined as

$$z_1 z_2 := (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1).$$

Theorem 1.1.3 (Complex Field):

The set of complex numbers $\mathbb{C}$ with the operations of addition and multiplication defined in **Definition 1.1.2** are *well-defined* and *satisfy the properties that we except them to have.*

Definition 1.1.4 (Imaginary Unit):

The **imaginary unit** i is defined as the complex number $(0, 1)$.

The imaginary unit i has a very interesting property, which is that

$$i^2 = (0, 1)(0, 1) = (-1, 0).$$

Corollary 1.1.5 (Decomposition of Complex):

A complex number $z = (x, y)$ can be written as $z = (x, 0) + i(y, 0)$.

Proof 1.1.5:

Consider a complex number $z = (x, 0) + i(y, 0)$. We have

$$z = (x, 0) + (0, 1)(y, 0) = (x, 0) + (0, y) = (x, y).$$

■

Theorem 1.1.6 (Embedding of Real in Complex):

Let R be the set of all complex numbers of the form $(x, 0)$, where $x \in \mathbb{R}$. That is

$$R = \{(x, 0) : x \in \mathbb{R}\}.$$

Then, R is a subset of $\mathbb{C}$ and R is *structurally equivalent* to $\mathbb{R}$.

Proof 1.1.6:

As we need to properly define the word *equivalent* before we can prove the theorem, we shall only first do an informal demonstration of the theorem here.

To show that R is *structurally equivalent* to $\mathbb{R}$, let's first show that R is *itself equivalent* to $\mathbb{R}$. Consider an element $(x, 0) \in R$, we can associate it to **a unique** element $x \in \mathbb{R}$ and **vice versa**. This is called a **bijection**, which somewhat means that R and $\mathbb{R}$ have the same number of elements.

Furthermore, we claim that the **addition** and **multiplication** of two elements in R are *equivalent* to the addition and multiplication of the corresponding elements in $\mathbb{R}$. To show this, consider two elements $(x_1, 0), (x_2, 0) \in R$, we have

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0),$$

which corresponds to the sum of x_1 and x_2 in $\mathbb{R}$. Similarly, we have

$$(x_1, 0)(x_2, 0) = (x_1 x_2, 0),$$

which also corresponds to the product of x_1 and x_2 in $\mathbb{R}$.

Since the **addition** and **multiplication** on these two sets are *equivalent*, any other operations derived from these two operations (such as subtraction, division, exponential) are also *equivalent*. Hence, we can conclude that R and $\mathbb{R}$ are *structurally equivalent*.

Convention 1.1.7:

Now, due to this *structural equivalence*, we will denote any complex number of the form $(x, 0)$ directly by x and **call** this number a **real number**.

Moreover, we will borrow any function $f : \mathbb{R} \mapsto \mathbb{R}$ to R, by defining a function $f_R : R \mapsto R$ as

$$f_R((x, 0)) := (f(x), 0).$$

Corollary 1.1.8 (Alternative Notation):

A complex number $z = (x, y)$ can be written as

$$z = (x, y) = (x, 0) + i(y, 0) = x + iy.$$

For a complex number $z = x + iy$, we call x the **real part** of z and denote it by $\mathfrak{R}(z)$ or $\mathrm{Re}(z)$, and we call y the **imaginary part** of z and denote it by $\mathfrak{I}(z)$ or $\mathrm{Im}(z)$.

Definition 1.1.9 (Complex Conjugate):

The **complex conjugate** of a complex number $z = x + iy$ is defined as

$$\bar{z} := x - iy.$$

One can immediately verify that the complex conjugate has the following property:

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2.$$

Due to **Convention 1.1.7**, we treat $x^2 + y^2$ as real number. Since $z\bar{z} = x^2 + y^2$ is a **non-negative real number**, we can take the square root of it.

Definition 1.1.10 (Modulus):

The **modulus** of a complex number $z = x + iy$ is defined as

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

Corollary 1.1.11 (Modulus and Conjugate):

The modulus of a complex number z is always a non-negative real number, that is,

$$|z| \geq 0.$$

Corollary 1.1.12 (Inverse of a Complex Number):

A complex number $z = x + iy \neq 0$ has a multiplicative inverse, which is given by

$$\frac{1}{z} = \frac{1}{z}\frac{\bar{z}}{\bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{x}{|z|} - i\frac{y}{|z|}$$

2. Polynomials and Complex Roots

Theorem 1.2.1 (Fundamental Theorem of Algebra):

Every non-constant polynomial with complex coefficients has at least one complex root.

Theorem 1.2.2 (Factorization of Polynomials):

If f is a degree n polynomial with complex coefficients, then f can be factored into n **linear factors**.

$$f = a(X - z_1)(X - z_2)...(X - z_n),$$

where $z_1, z_2, ..., z_n$ are the complex roots of f and a is the leading coefficient of f.

3. Geometric Interpretation of Complex Numbers

Lemma 1.3.1 (Euler's Formula):

Let $\theta \in \mathbb{R}$. Then, we have

$$e^{i\theta} = \cos(\theta) + i\sin(\theta).$$

Corollary 1.3.2 (Modulus of Exponential):

One can immediately verify that the modulus of $e^{i\theta}$ is always 1, that is,

$$\left|e^{i\theta}\right| = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = 1.$$

Corollary 1.3.3 (Polar Form):

One can also verify that any complex number z can be written in the form

$$z = re^{i\theta},$$

where $r = |z| \in \mathbb{R}$ and $\arg(z) := \theta$ is called the **argument** of z.

4. Properties of Complex Numbers

Definition 1.4.1 (Purely Imaginary Numbers):

A complex number z is called a **purely imaginary number** if its real part is zero, that is,

$$\mathfrak{R}(z) = 0.$$

Corollary 1.4.2 (Re and Im Function):

The real part and the imaginary part of a complex number z can be expressed in terms of z and its complex conjugate as follows:

$$\mathfrak{R}(z) = \frac{z + \bar{z}}{2} \text{ and } \mathfrak{I}(z) = \frac{z - \bar{z}}{2i}.$$

Corollary 1.4.3 (Conjugate, Real, Purely Imaginary):

If a complex number z has the following property:

$$z = -\bar{z},$$

then it is a purely imaginary number.

If a complex number z has the following property:

$$z = \bar{z},$$

then it is a real number.

Chapter 2. Algebraic Structures

1. Groups, Rings, and Fields

Definition 2.1.1 (Groups):

A **group** is a set G together with a binary operation $\cdot$ that satisfies the following properties:

1. **Closure**: For all $a, b \in G$, we have $a \cdot b \in G$.
2. **Associativity**: For all $a, b, c \in G$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity**: There exists an element $e \in G$ such that for all $a \in G$, we have $e \cdot a = a \cdot e = a$.
4. **Inverse**: For each element $a \in G$, there exists an element $b \in G$ such that $a \cdot b = b \cdot a = e$.

Definition 2.1.2 (Monoids and Semigroups):

A **monoid** is a set S together with a binary operation $\cdot$ that satisfies the following properties:

1. **Closure**
2. **Associativity**
3. **Identity**

A **semigroup** is a set S together with a binary operation $\cdot$ that satisfies the following properties:

1. **Closure**
2. **Associativity**

Corollary 2.1.3 (Uniqueness of Identity and Inverse):

The **identity element** and the **inverse of each element** in a group are unique.

Proof 2.1.3:

Let G be a group and let e and e' be two identity elements in G. Then, we have

$$e = e \cdot e' = e'.$$

Hence, the identity element is unique.

Let $a \in G$ and let b and b' be two inverses of a. Then, we have

$$b = b \cdot e = b \cdot (a \cdot b') = (b \cdot a) \cdot b' = e \cdot b' = b'.$$

Hence, the inverse of each element is unique.

■

Definition 2.1.4 (Abeliang Groups):

A group G is called an **Abelian group** if the binary operation $\cdot$ is **commutative**, that is, for all $a, b \in G$, we have $a \cdot b = b \cdot a$.

Definition 2.1.5 (Rings):

A **ring** is a set R together with two binary operations $+$, called **addition**, and $\cdot$, called **multiplication**, such that:

1. $(R, +)$ is an **Abelian group**.
2. $(R, \cdot)$ is a **monoid**.
3. **Distributive Laws**: For all $a, b, c \in R$, we have
   - $a \cdot (b + c) = a \cdot b + a \cdot c$
   - $(a + b) \cdot c = a \cdot c + b \cdot c$

Definition 2.1.6 (Commutative Rings):

A **commutative ring** is a ring $R(+, \cdot)$ such that the **multiplication is commutative**, that is, for all $a, b \in R$, we have $a \cdot b = b \cdot a$.

Definition 2.1.7 (Fields):

A **field** is a set F together with two binary operations $+$ and $\cdot$, with its **additive identity** denoted by 0, such that:

1. $F(+, \cdot)$ is a **commutative ring**.
2. Every element $f \in F$ such that $f \neq 0$ has a **multiplicative inverse** in F.

Theorem 2.1.8 (Real Field):

The set of real numbers $\mathbb{R}$ with the usual addition and multiplication forms a **field**.

Proof 2.1.8:

Let's start by showing that $\mathbb{R}(+)$ is an Abelian group.

1. **Closure**: For all $a, b \in \mathbb{R}$, we have $a + b \in \mathbb{R}$.
2. **Associativity**: For all $a, b, c \in \mathbb{R}$, we have $(a + b) + c = a + (b + c)$.
3. **Identity**: The additive identity is 0, since for all $a \in \mathbb{R}$, we have $0 + a = a + 0 = a$.
4. **Inverse**: For each element $a \in \mathbb{R}$, the additive inverse is $-a$, since we have $a + (-a) = (-a) + a = 0$.
5. **Commutativity**: For all $a, b \in \mathbb{R}$, we have $a + b = b + a$.

Now, let's show that $\mathbb{R}(\cdot)$ is a monoid.

1. **Closure**: For all $a, b \in \mathbb{R}$, we have $a \cdot b \in \mathbb{R}$.
2. **Associativity**: For all $a, b, c \in \mathbb{R}$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity**: The multiplicative identity is 1, since for all $a \in \mathbb{R}$, we have $1 \cdot a = a \cdot 1 = a$.

Due to the **distributive laws** of real numbers, we can conclude that $\mathbb{R}$ is a **ring**. Moreover, it is a **commutative ring** since the multiplication of real numbers is commutative. Finally, **every non-zero real number has a multiplicative inverse**, which is also a real number. Hence, we can conclude that $\mathbb{R}$ is a **field**.

$\blacksquare$

Theorem 2.1.9 (Complex Field):

Now, we can formally restate **Theorem 1.1.3** as follows: The set of complex numbers $\mathbb{C}$ with the operations of addition and multiplication defined in **Definition 1.1.2** forms a **field**.

The proof of this theorem follows the same idea as the proof for **Theorem 2.1.8**.

2. Isomorphisms

Two sets composed of very different mathematical objects might be *structurally similar* to each other to a point that an theorem proven in one can be directly applied to the other. Identifying this *similarity* is particularly useful, as it allows us to freely switch different perspectives and apply the tools we have in one to another. This is the motivation behind the concept of **isomorphism**.

Definition 2.2.1 (Isomorphisms):

An **isomorphism** is a **structure-preserving bijection** between two **structures**. If such an isomorphism exists between two structures, we say that these two structures are **isomorphic** to each other.

Definition 2.2.2 (Isomorphism between Algebraic Structures):

Let A and B be two algebraic structures of the same type (for example, two groups, or two rings). An **isomorphism** between A and B is a bijection $f : A \mapsto B$ such that for all elements $a_1, a_2 \in A$ and for all binary operation $\cdot_A$ defined on A, we have

$$f(a_1 \cdot_A a_2) = f(a_1) \cdot_B f(a_2),$$

where $\cdot_B$ is the corresponding binary operation defined on B.

Theorem 2.2.3 (Isomorphism between $\mathbb{R}$ and R):

Now we can formally define the meaning of *structural equivalence* in **Theorem 1.1.6** as follows: The **field** of real numbers $\mathbb{R}$ is isomorphic to the **field** form by complex numbers of the form $(x, 0)$, where $x \in \mathbb{R}$.

Proof 2.2.3:

The proof is surprisingly simple and is already outlined in **Proof 1.1.6**. One only needs to formally reformulate the proof in **Proof 1.1.6** using the definition of isomorphism between algebraic structures.

Hence, we can conclude that $\mathbb{R}$ and R are isomorphic to each other.

Chapter 3. Vector Spaces

1. Structure of Vector Spaces

Definition 3.1.1 (Vector Spaces):

A **vector space** over a field F is a set V together with two operations, called **vector addition** and **scalar multiplication**, such that:

1. V and $+$, the **vector addition**, together form an **Abelian group**, with the **additive identity** denoted by 0.
2. The **scalar multiplication** is a function $F \times V \mapsto V$ that takes a scalar $a \in F$ and a vector $v \in V$ and produces a new vector $av \in V$ that satisfies the following properties:
   - **Closure**: For all $a \in F$ and $v \in V$, we have $av \in V$.
   - **Associativity**: For all $a, b \in F$ and $v \in V$, we have $(ab)v = a(bv)$.
   - **Identity**: For all $v \in V$, we have $1v = v$, where 1 is the **multiplicative identity** in F.
   - **Distributive Laws**: For all $a, b \in F$ and $u, v \in V$, we have
     1. $a(u + v) = au + av$
     2. $(a + b)v = av + bv$

Definition 3.1.2 (Linear Combination):

A **linear combination** of a set of vectors $S \subseteq V$ is a vector that can be expressed as **a finite sum** of scalar multiples of vectors in S. That is, a vector $v \in V$ is a linear combination of S if there exist scalars $\lambda^1, \cdots, \lambda^n \in F$ and vectors $s_1, \cdots, s_n \in S$ such that

$$v = \sum_{i=1}^{n} \lambda^i s_i.$$

Definition 3.1.3 (Span):

The **span** of a set of vectors $S \subseteq V$ is the set of all linear combinations of S. That is, the span of S is defined as

$$\operatorname{span}_F(S) := \left\{ \sum_{i=1}^{n} \lambda^i s_i : n \in \mathbb{N}, \lambda^1, \cdots, \lambda^n \in F, s_1, \cdots, s_n \in S \right\}.$$

Corollary 3.1.4:

The span of a set of vectors S always contains the zero vector.

Definition 3.1.5 (Linear Independence):

A set of vectors $S \subseteq V$ is said to be **linear independent** if the only linear combination of S that equals the zero vector is the **trivial linear combination**, that is, the linear combination where all scalars are zero. Formally, S is linear independent if for any scalars $\lambda^1, \cdots, \lambda^n \in F$ and vectors $s_1, \cdots, s_n \in S$, we have

$$\sum_{i=1}^{n} \lambda^i s_i = 0 \Leftrightarrow \lambda^1 = \lambda^2 = \ldots = \lambda^n = 0.$$

Definition 3.1.6 (Basis):

A **basis** of a vector space V is an **ordered list** of vectors $B \subseteq V$ that is both **linearly independent** and **spans** V. That is, B is a basis of V if

1. B is linear independent.
2. $\text{span}(B)$ = V.

Definition 3.1.7 (Subspace):

A **subspace** of a vector space V is a subset $W \subseteq V$ that is itself a vector space with the same field under the same operations of vector addition and scalar multiplication defined on V.

Theorem 3.1.8 (Subspace Test):

A subset $W \subseteq V$ is a subspace of a vector space V if and only if for all $u, v \in W$ and $a \in F$, we have

$$au + v \in W.$$

Proof 3.1.8:

Let's first show that $(W, +)$ is an Abelian group.

1. **Closure**: Let $a = 1$, this means that for all $u, v \in W$, $u + w \in W$. This implies that W is closed under vector addition.
2. **Associativity**: Since $W \subseteq V$ and $(V, +)$ is an Abelian group, we have for all $u, v, w \in W$, $(u + v) + w = u + (v + w)$.
3. **Identity**: Let $a = -1$ and $u \in W$. Since $-1u + u = 0 \in W$, W contains the additive identity.
4. **Inverse**: Let $a = -1$ and $v = 0$. Since $-1u + 0 = -u \in W$, W contains the additive inverse of each of its elements.
5. **Commutativity**: Since $W \subseteq V$ and $(V, +)$ is an Abelian group, we have for all $u, v \in W$, $u + v = v + u$.

Then, let's show that the scalar multiplication on W satisfies the properties of scalar multiplication on a vector space.

1. **Closure**: Let $v = 0$, then for all $a \in F$ and $u \in W$, we have $au + 0 = au \in W$. This implies that W is closed under scalar multiplication.
2. **Associativity**, **Identity**, and **Distributive Laws**: Since $W \subseteq V$ and the scalar multiplication on V satisfies these properties, we have for all $a, b \in F$ and $u, v \in W$, $(ab)u = a(bu)$, $1u = u$, $a(u + v) = au + av$, and $(a + b)v = av + bv$.

This shows that if for all $u, v \in W$ and $a \in F$, we have $au + v \in W$, then W is a subspace of V.

Now, let's show the converse. If W is a subspace of V, then W is closed under vector addition and scalar multiplication. Hence, for all $u, v \in W$ and $a \in F$, we have $au + v \in W$.

■

Theorem 3.1.9 (Intersection of Subspaces):

The intersection of any collection of subspaces of a vector space V is also a subspace of V.

Proof 3.1.9:

Let S be a collection of subspaces of V and let $W = \bigcap_{U \in S} U$. We need to show that W is a subspace of V.

Let $u, v \in W$ and $a \in F$. Since $u, v \in W$, we have for all $U \in S$, $u, v \in U$.

Since each U is a subspace of V, we have for all $U \in S$, $au + v \in U$. Hence, we have for all $U \in S$, $au + v \in U$, which implies that $au + v \in W$. This shows that W is a subspace of V.

■

Theorem 3.1.10 (Span is a Subspace):

Let W be a subset of a vector space V. Then, the span of W is a subspace of V.

Proof 3.1.10:

Let $u, v \in \operatorname{span}(W)$ and $a \in F$. Since $u, v \in \operatorname{span}(W)$, we have that there exist scalars $\lambda^1, \cdots, \lambda^n \in F$ and vectors $s_1, \cdots, s_n \in W$ such that

$$u = \sum_{i=1}^{n} \lambda^i s_i.$$

Similarly, there exist scalars $\mu^1, \cdots, \mu^m \in F$ and vectors $t_1, \cdots, t_m \in W$ such that

$$v = \sum_{j=1}^{m} \mu^j t_j.$$

Hence, we have

$$au + v = a \sum_{i=1}^{n} \lambda^i s_i + \sum_{j=1}^{m} \mu^j t_j = \sum_{i=1}^{n} (a\lambda^i) s_i + \sum_{j=1}^{m} \mu^j t_j,$$

which is a linear combination of vectors in W. This shows that $au + v \in \operatorname{span}(W)$, which implies that $\operatorname{span}(W)$ is a subspace of V.

■

Theorem 3.1.11 (Span of a Subspace):

The span of a subspace W of a vector space V is equal to W itself, that is, $\operatorname{span}(W) = W$.

Proof 3.1.11:

Since W is a subspace of V, it is closed under linear combinations. Hence, we have $\operatorname{span}(W) \subseteq W$. On the other hand, since $W \subseteq W$, we have $W \subseteq \operatorname{span}(W)$. This shows that $\operatorname{span}(W) = W$.

■

Theorem 3.1.12 (Span of Subsets):

Let W be a subset of a vector space V. Then, the span of W is the smallest subspace of V that contains W. In other words, let S be the collection of all subspaces of V that contain W, then

$$\operatorname{span}(W) = \bigcap_{U \in S} U.$$

Proof 3.1.12:

Let's start by first showing that for any $U \in S$,

$$\operatorname{span}(W) \subseteq U.$$

Since $W \subseteq U$, we have that any linear combination of vectors in W is also a linear combination of vectors in U. Hence, we have $\operatorname{span}(W) \subseteq \operatorname{span}(U)$, but since U is a subspace of V, we have $\operatorname{span}(U) = U$. This shows that $\operatorname{span}(W) \subseteq U$ for any $U \in S$. Therefore,

$$\operatorname{span}(W) \subseteq \bigcap_{U \in S} U.$$

However, since $\operatorname{span}(W)$ is a subspace of V that contains W, we have $\operatorname{span}(W) \in S$. Hence, we have

$$\bigcap_{U \in S} U \subseteq \operatorname{span}(W).$$

Thus, this shows that $\operatorname{span}(W) = \bigcap_{U \in S} U$.

Definition 3.1.13 (Linear Transformations):

Let V and W be vector spaces over a field F. A function $T : V \mapsto W$ is called a **linear transformation** if for all $u, v \in V$ and $a \in F$, we have

$$T(au + v) = aT(u) + T(v).$$

Definition 3.1.14 (Isomorphism between Vector Spaces):

Let V and W be two vector spaces over the same field F. An **isomorphism** between V and W is a **bijective linear transformation** $T : V \mapsto W$. If such an isomorphism exists, we say that V and W are **isomorphic** to each other.

2. Cartesian Space

Definition 3.2.1 (n-tuples):

An **n-tuple** is an ordered list of n elements, where n is a positive integer. An n-tuple, x can be denoted by

$$x = \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix},$$

where $x^1, x^2, ..., x^n$ are the elements of the n-tuple.

Definition 3.2.2 (Cartesian Spaces: n-tuples Vector Spaces):

Let F be a field and let n be a positive integer. The set of all n-tuples of elements in F, denoted by F^n, forms a vector space over F with **addition** and **scalar multiplication** defined as follows:

1. **Addition**:

   Let $x = \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix}$ and $y = \begin{pmatrix} y^1 \\ \vdots \\ y^n \end{pmatrix}$ be two n-tuples in F^n. Then, the sum of x and y is defined as

$$x + y = \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix} + \begin{pmatrix} y^1 \\ \vdots \\ y^n \end{pmatrix} := \begin{pmatrix} x^1 + y^1 \\ \vdots \\ x^n + y^n \end{pmatrix}.$$

2. **Scalar Multiplication**:

   Let $x = \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix}$ be an n-tuple in F^n and let $\lambda \in F$ be a scalar. Then, the scalar multiplication of a and x is defined as

$$\lambda x = \lambda \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix} := \begin{pmatrix} \lambda x^1 \\ \vdots \\ \lambda x^n \end{pmatrix}.$$

The zero vector in F^n is the n-tuple where all components are zero, that is, $\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$.

Proof 3.2.2:

Since we are defining the addition and scalar multiplication component-wise, we can verify that the vector space axioms are satisfied by directly applying the corresponding properties of the field F to each component of the n-tuples. Hence, we can conclude that F^n is a vector space over F.

■

Theorem 3.2.3 (Standard Basis):

Let F^n be a vector space of n-tuples over a field F. Then, the set of n-tuples

$$\mathfrak{B} = \left\{ e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \cdots, e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}$$

is a basis of F^n, which is called the **standard basis** of F^n.

Proof 3.2.3:

Let's first show that $\mathfrak{B}$ is linear independent. Consider a linear combination of $\mathfrak{B}$ that equals the zero vector, that is,

$$\sum_{i=1}^{n} \lambda^i e_i = 0.$$

But

$$\sum_{i=1}^{n} \lambda^i e_i = \begin{pmatrix} \lambda^1 \\ \lambda^2 \\ \vdots \\ \lambda^n \end{pmatrix}.$$

Hence, the only way for this linear combination to equal the zero vector is that $\lambda^1 = \lambda^2 = \ldots = \lambda^n = 0$. This shows that $\mathfrak{B}$ is linear independent.

Now, let's show that $\mathfrak{B}$ spans F^n. Consider an arbitrary n-tuple $x = \begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix} \in F^n$. We can write x as a linear combination of the vectors in $\mathfrak{B}$ as follows:

$$x = \sum_{i=1}^{n} x^i e_i.$$

That is $F^n \subseteq \operatorname{span}(\mathfrak{B})$. On the other hand, since F^n is a vector space, it is closed under linear combinations. Hence, we have $\operatorname{span}(\mathfrak{B}) \subseteq F^n$. This shows that $\operatorname{span}(\mathfrak{B}) = F^n$.

■

Corollary 3.2.4 (Decomposition of Linear Transformations):

Let V and W be two Cartesian spaces over the same field F, that is, $V = F^m$ and $W = F^n$ for some positive integers m and n, with standard basis $\{\mathscr{v}_1, \cdots, \mathscr{v}_m\}$ and $\{\mathscr{w}_1, \cdots, \mathscr{w}_n\}$ respectively. Consider a linear transformation $T : V \mapsto W$, we have

$$T(v) = T\left(\sum_{i=1}^{m} v^i \mathscr{v}_i\right) = \sum_{i=1}^{m} v^i T(\mathscr{v}_i),$$

where $v^i \in F$ is the i-th components of v with respect to the standard basis of V. Now, since $T(\mathscr{v}_i)$ is a vector in W, we can write it as a linear combination of the standard basis of W as follows:

$$T(\mathscr{v}_i) = \sum_{j=1}^{n} T_i^j \mathscr{w}_j,$$

where $T_i^j \in F$ is the j-th component of $T(\mathscr{v}_i)$ with respect to the standard basis of W. Hence, we have

$$T(v) = \sum_{i=1}^{m} v^i \left(\sum_{j=1}^{n} T_i^j w_j \right) = \sum_{j=1}^{n} \sum_{i=1}^{m} T_i^j v^i w_j.$$

This means that **any linear transformation** between Cartesian spaces **can define and be defined by a set of scalars** $T_i^j \in F$ for $i = 1, \cdots, m$ and $j = 1, \cdots, n$.

Convention 3.2.5 (Matrix Representation of Linear Transformations):

In order to conveniently represent a linear transformation between Cartesian spaces, we can arrange the scalars T_i^j into a table like object called **matrix** as follows:

$$\boldsymbol{T} = \begin{pmatrix} T_1^1 & T_2^1 & \cdots & T_m^1 \\ T_1^2 & T_2^2 & \cdots & T_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ T_1^n & T_2^n & \cdots & T_m^n \end{pmatrix}$$

Applying the linear transformation T to a vector $v \in V$ is represented as follows:

$$T(v) = \boldsymbol{T}v = \begin{pmatrix} T_1^1 & T_2^1 & \cdots & T_m^1 \\ T_1^2 & T_2^2 & \cdots & T_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ T_1^n & T_2^n & \cdots & T_m^n \end{pmatrix} \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^m \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^m T_i^1 v^i \\ \sum_{i=1}^m T_i^2 v^i \\ \vdots \\ \sum_{i=1}^m T_i^n v^i \end{pmatrix} = \sum_{j=1}^{n} \sum_{i=1}^{m} T_i^j v^i w_j.$$

In particular, if the matrix $\boldsymbol{T}$ has n rows and m columns, we say that $\boldsymbol{T}$ is an **n by m matrix** or $\boldsymbol{n \times m}$ **matrix**.

The operation of applying a linear transformation involves putting a matrix and a vector side by side and performing a series of multiplications and additions to produce a new vector. This operation is called **matrix multiplication**. We shall formally define this operation.

Definition 3.2.6 (Matrix Multiplication I):

Let $T : F^m \mapsto F^n$ be a linear transformation represented by the matrix

$$\boldsymbol{T} = \begin{pmatrix} T_1^1 & T_2^1 & \cdots & T_m^1 \\ T_1^2 & T_2^2 & \cdots & T_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ T_1^n & T_2^n & \cdots & T_m^n \end{pmatrix}$$

and let $v \in F^m$ be a vector represented by the n-tuple

$$v = \begin{pmatrix} v^1 \\ v^2 \\ \vdots \\ v^m \end{pmatrix}.$$

Then, the **matrix multiplication** of T and v is defined as

$$\boldsymbol{T}v := \begin{pmatrix} \sum_{i=1}^m T_i^1 v^i \\ \sum_{i=1}^m T_i^2 v^i \\ \vdots \\ \sum_{i=1}^m T_i^n v^i \end{pmatrix}.$$

Theorem 3.2.7 (Matrices and Linear Transformations):

Let $\mathcal{T}$ be the set of all linear transformations from F^m to F^n and let $M_{n\times m}(F)$ represents all linear transformations from F^m to F^n defined by n by m matrices with entries in F. Then, there is

$$\mathcal{T} = M_{n\times m}(F).$$

Proof 3.2.7:

Let $T : F^m \mapsto F^n$ be a linear transformation. Then, we can represent T by a matrix $\boldsymbol{T}$ as shown in **Corollary 3.2.4** and **Convention 3.2.5**. Hence, we have $T \in M_{n\times m}(F)$, which implies that $\mathcal{T} \subseteq M_{n\times m}(F)$.

Now, let $\boldsymbol{T}$ be an arbitrary n by m matrix with entries in F. Again, as shown in **Corollary 3.2.4** and **Convention 3.2.5** we can define a linear transformation $T : F^m \mapsto F^n$ by

$$T(v) = \boldsymbol{T}v$$

for all $v \in F^m$. This shows that for any matrix in $M_{n\times m}(F)$, there is a corresponding linear transformation in $\mathcal{T}$. Hence, we have $M_{n\times m}(F) \subseteq \mathcal{T}$.

Therefore, we can conclude that $\mathcal{T} = M_{n\times m}(F)$.

■

Theorem 3.2.8 (Matrices are Vectors):

Let $M_{n\times m}(F)$ be the set of all n by m matrices with entries in a field F. Then, $M_{n\times m}(F)$ is a vector space over F with **addition** and **scalar multiplication** defined as follows:

$$\boldsymbol{A}+\boldsymbol{B} = \begin{pmatrix} A_1^1 & A_2^1 & \cdots & A_m^1 \\ A_1^2 & A_2^2 & \cdots & A_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ A_1^n & A_2^n & \cdots & A_m^n \end{pmatrix} + \begin{pmatrix} B_1^1 & B_2^1 & \cdots & B_m^1 \\ B_1^2 & B_2^2 & \cdots & B_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ B_1^n & B_2^n & \cdots & B_m^n \end{pmatrix} := \begin{pmatrix} A_1^1+B_1^1 & A_2^1+B_2^1 & \cdots & A_m^1+B_m^1 \\ A_1^2+B_1^2 & A_2^2+B_2^2 & \cdots & A_m^2+B_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ A_1^n+B_1^n & A_2^n+B_2^n & \cdots & A_m^n+B_m^n \end{pmatrix},$$

$$a\boldsymbol{A} = a\begin{pmatrix} A_1^1 & A_2^1 & \cdots & A_m^1 \\ A_1^2 & A_2^2 & \cdots & A_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ A_1^n & A_2^n & \cdots & A_m^n \end{pmatrix} := \begin{pmatrix} aA_1^1 & aA_2^1 & \cdots & aA_m^1 \\ aA_1^2 & aA_2^2 & \cdots & aA_m^2 \\ \vdots & \vdots & \ddots & \vdots \\ aA_1^n & aA_2^n & \cdots & aA_m^n \end{pmatrix}.$$

Proof 3.2.8:

It is straightforward to verify that the vector space axioms are satisfied by directly applying the corresponding properties of the field F to each component of the matrices. Hence, we can conclude that $M_{n\times m}(F)$ is a vector space over F.

■

In some cases, we need to perform multiple linear transformations in a sequence. For example, let $T : F^m \mapsto F^n$ and $S : F^n \mapsto F^p$ be two linear transformations, we might want to apply T to a vector $v \in F^m$ first and then apply S to the result, that is,

$$S(T(v)) = S\left(\begin{pmatrix} \sum_{i=1}^m T_i^1 v^i \\ \sum_{i=1}^m T_i^2 v^i \\ \vdots \\ \sum_{i=1}^m T_i^n v^i \end{pmatrix}\right) = \begin{pmatrix} \sum_{j=1}^n \sum_{i=1}^m S_j^1 T_i^j v^i \\ \sum_{j=1}^n \sum_{i=1}^m S_j^2 T_i^j v^i \\ \vdots \\ \sum_{j=1}^n \sum_{i=1}^m S_j^p T_i^j v^i \end{pmatrix}$$

Notice how we can first compute the sum for the index j, that is

$$S(T(v)) = \begin{pmatrix} \sum_{i=1}^m \left(\sum_{j=1}^n S_j^1 T_i^j\right) v^i \\ \sum_{i=1}^m \left(\sum_{j=1}^n S_j^2 T_i^j\right) v^i \\ \vdots \\ \sum_{i=1}^m \left(\sum_{j=1}^n S_j^p T_i^j\right) v^i \end{pmatrix}.$$

Since the result of the sum for the index j is a scalar, $S(T(v))$ uniquely defines a linear transformation $R : F^m \mapsto F^p$ by the matrix $\boldsymbol{R}$:

$$\boldsymbol{R} = \begin{pmatrix} \sum_{j=1}^n S_j^1 T_1^j & \sum_{j=1}^n S_j^1 T_2^j & \cdots & \sum_{j=1}^n S_j^1 T_m^j \\ \sum_{j=1}^n S_j^2 T_1^j & \sum_{j=1}^n S_j^2 T_2^j & \cdots & \sum_{j=1}^n S_j^2 T_m^j \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^n S_j^p T_1^j & \sum_{j=1}^n S_j^p T_2^j & \cdots & \sum_{j=1}^n S_j^p T_m^j \end{pmatrix}.$$

This is particularly convenient, as now we have

$$S(T(v)) = \boldsymbol{S}(\boldsymbol{T}v) = \boldsymbol{R}v.$$

Hence, for simplicity, we will denote the matrix $\boldsymbol{R}$ as a **product** of the matrices $\boldsymbol{S}$ and $\boldsymbol{T}$ to mimic the associativity. That is,

$$\boldsymbol{R} = \boldsymbol{S}\boldsymbol{T}.$$

In other words,

$$\boldsymbol{S}(\boldsymbol{T}v) = (\boldsymbol{S}\boldsymbol{T})v = \boldsymbol{R}v.$$

Definition 3.2.9 (Matrix Multiplication II):

Let $\boldsymbol{S}$ be an p by n matrix and let $\boldsymbol{T}$ be an n by m matrix. Then, the **matrix multiplication** of $\boldsymbol{S}$ and $\boldsymbol{T}$ is defined as the p by m matrix

$$\boldsymbol{S}\boldsymbol{T} = \begin{pmatrix} S_1^1 & \cdots & S_n^1 \\ \vdots & \ddots & \vdots \\ S_1^p & \cdots & S_n^p \end{pmatrix} \begin{pmatrix} T_1^1 & \cdots & T_m^1 \\ \vdots & \ddots & \vdots \\ T_1^n & \cdots & T_m^n \end{pmatrix} := \begin{pmatrix} \sum_{j=1}^n S_j^1 T_1^j & \cdots & \sum_{j=1}^n S_j^1 T_m^j \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^n S_j^p T_1^j & \cdots & \sum_{j=1}^n S_j^p T_m^j \end{pmatrix}.$$

Corollary 3.2.10 (Associativity of Matrix Multiplication):

Let $\boldsymbol{R}$, $\boldsymbol{S}$, and $\boldsymbol{T}$ be matrices such that the products $\boldsymbol{RS}$ and $\boldsymbol{ST}$ are defined. Then, we have

$$\boldsymbol{R}(\boldsymbol{ST}) = (\boldsymbol{RS})\boldsymbol{T}.$$

Proof 3.2.10:

Applying $\boldsymbol{R}(\boldsymbol{ST})$ on an arbitrary vector v, since $(\boldsymbol{ST})v = \boldsymbol{S}(\boldsymbol{T}v)$ we have

$$\boldsymbol{R}(\boldsymbol{ST})v = \boldsymbol{R}(\boldsymbol{S}(\boldsymbol{T}v)),$$

but $\boldsymbol{T}v$ is just a vector, hence we have

$$\boldsymbol{R}(\boldsymbol{S}(\boldsymbol{T}v)) = (\boldsymbol{RS})(\boldsymbol{T}v) = (\boldsymbol{RS})\boldsymbol{T}v.$$

This shows that $\boldsymbol{R}(\boldsymbol{ST}) = (\boldsymbol{RS})\boldsymbol{T}$.

$\blacksquare$

3. Linear Equations

Definition 3.3.1 (Linear Equations):

A **linear equation** is an equation of the form

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = b,$$

where $a_1, a_2, \cdots, a_n$ and b are constants and $x_1, x_2, \cdots, x_n$ are variables.

Corollary 3.3.2 (Matrix Form of Linear Equations):

A system of linear equations can be represented in matrix form as follows:

$$\boldsymbol{A}x = \boldsymbol{A}\begin{pmatrix} x^1 \\ x^2 \\ \vdots \\ x^n \end{pmatrix} = \begin{pmatrix} b^1 \\ b^2 \\ \vdots \\ b^m \end{pmatrix} = b.$$

where $\boldsymbol{A}$ is an m by n matrix of coefficients, x is a vector of variables, and b is a vector of constants.